# CO1 AT-2

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Fundamentals of Data Science

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# STUDENT PERFORMANCE ANALYSIS

## Data Science Process: From Data Collection to Data Transformation

### Data Collection (Raw data)

- Attendance records
  - Present / Absent
  - Percentage
- Internal marks
  - Test scores
  - Subject-wise marks
- Assignment Scores
  - Assignment 1, 2, ---
  - marks
- Final exam results
  - Subject-wise marks
  - Total marks

### Data Integration

- Attendance data
- Internal marks
- Assignment scores
- Finalexam results

→ Student Performance dataset

- Student ID
- Subject
- Attendance
- Internal marks
- Assignment marks
- final exam marks

### Data Cleaning

- Handle missing values (e.g., absent marks)
- Remove duplicate records (same student entry)
- Fix inconsistent formats (e.g., date, roll number)
- validate data types (numeric / categorical)
- Check outliers (invalid marks or attendance)

### Data Transformation

- Encode categorical data (e.g., subject codes)
- Create new features (e.g., total marks, avg marks)
- Normalize/Scale values (e.g., marks to 0-1 range)
- Aggregate data (e.g. subject-wise or student-wise summary)
- Convert to analysis-ready format

### Analysis Ready Dataset



- cleaned and transformed data
- Ready for analysis / model building
- can be used for performance evaluation, visualization and Prediction.

# HOSPITAL DATA COLLECTION AND ANALYSIS

## Data Sources and their Use in Analysis

Data Sources (In Hospital)

### 1. Registration Forms
- Patient ID
- Name, Age, Gender
- Contact details
- Address
- visit date

### 2. Lab Reports
- Blood test results
- Urine test results
- Imaging reports
- Test dates
- lab technician notes

### 3. Medical Devices
- Vital signs (BP, Heart rate temp)
- Continuous monitoring data (ICU, wards)
- Device timestamps

### 4. Doctor Notes
- Symptoms
- Diagnosis
- Treatment plan
- Prescriptions
- follow-up notes

### 5. Data Integration
- Combine data from all sources
- match using Patient ID
- Align different formats
- Create a single dataset

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### Integrated Hospital dataset
- Patient ID
- Demographics (age, gender)
- Lab results
- Vital signs
- Doctor notes
- Treatment info

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### 6. Data Preprocessing
- Handle missing values
- Remove duplicate records
- standardize data formats
- Convert data types
- extract useful features (e.g., latest test result)

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### Cleaned Dataset
- Consistent format
- No missing / duplicate data
- Ready for analysis

### 7. Analysis and usage

#### Disease Diagnosis support
- Analyze symptoms and test results
- Identify possible diseases

#### Patient Monitoring
- Track Vital Signs
- Detect abnormal patterns
- early warning for Critical conditions

#### Treatment Analysis
- Evaluate treatment effectiveness
- Compare outcomes
- support personalized treatment plans

#### Reporting and Research
- Generate health reports.
- Analyze patient trends.
- supporting medical research

# Data Cleaning for E-Commerce Dataset

Handle missing values, Duplicates, wrong Entries and outliers

## 1. Raw Dataset

- Customer details (ID, name, contacts)
- order information (order ID, date)
- Product details (Product ID, name, Price)
- Purchase amounts (quantity, total amount)

## 2. Common Data Quality Issues

| Missing customer details | Duplicate orders | Wrong product entries | Unusual Purchase amounts |
|---|---|---|---|
| • missing name<br>• missing contact<br>• missing address | • Same order ID<br>• Repeated entries<br>• multiple records for same order | • invalid product ID<br>• Incorrect Product name<br>• wrong category or price | • very high amounts<br>• very low amounts<br>• possible errors or fraud |

## 3. Data cleaning steps

1. Handle missing values
2. Remove duplicate rows
3. Correct wrong product entries
4. Detect and treat outliers
5. Standardize formats

## 4. cleaning Process

### Identify issues

- checking missing values
- find duplicates
- validate product details
- Detect unusual amounts

### Apply Cleaning techniques

- Impute or remove missing
- Drop duplicate orders
- Correct product entries
- Handle outliers
- standardize data formats

### Verify cleaned data

- Recheck for missing values
- confirm no duplicates
- Validate Product details

### Create consistent data set

- cleaned customer records
- valid Product information
- correct order amounts

## 5. Cleaned Dataset (Ready for analysis)

- complete customer details
- unique orders
- Correct product info
- Reasonable Purchase amounts

## 6. Uses in Analysis

### Customer Analysis

- Customer segmentation
- Purchase behaviour
- Identify high-value customers

### Product Analysis

- TOP selling products
- Category wise sales
- Predict Performance

### Sales Analysis

- Sales trends
- monthly / seasonal analysis
- Revenue forecasting

### fraud Detection

- Detect suspicious orders
- Identify unusual transactions,
- Prevent fraudulent activities

# SMART CITY DATA INTEGRATION

from multiple sources to a Single Dataset

## 1. Data Sources

### CCTV Cameras
- vehicle count
- Traffic flow
- Image/video data
- Time and location

### GPS Devices
- vehicle location
- Speed
- Route information
- Timestamp

### Sensors
- Traffic volume
- Road occupancy
- speed detection
- environmental data (eg., air quality, noise)

### Mobile Applications
- User location
- Traffic reports
- Crowd-sourced data
- Incident information
- Timestamp

## 2. Data Integration Process

### 1. Data Collection
- Collect data from all sources
- store in separate files (video logs, GPS logs, sensor data)

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### 2. Data Standardization
- Convert to common format
- standardize datatypes
- use consistent units (eg., time, location)
- clean missing or invalid values

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### 3. Data matching & merging
- use common attributes (eg., timestamp, location)
- Align data from different sources
- merge into a single dataset

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### 4. Integrated dataset
- Combined traffic data
- Includes vehicle count, speed, location, sensor reading, app reports.
- Unified format
- Ready for analysis

## 3. Uses in Analysis (Smart City Applications)

### Traffic Management
- monitor real time traffic
- Detect congestion
- optimize traffic signals

### URBAN Planning
- Analyze traffic Patterns
- Plan road infrastructure
- Identify high-traffic areas

### Incident Detection
- Detect accidents
- Identify abnormal traffic
- Enable quick response

### Environmental monitoring
- Analyze air quality & noise
- Study environmental impact
- support sustainable planning.

# CUSTOMER DATA TRANSFORMATION for LOAN APPROVAL PREDICTION

· Prepare data using Encoding, Normalization and Feature Scaling

## 1. Raw Customer Data

| Attribute | Description |
|---|---|
| Income | Annual Income (numeric) |
| Age | Age in years (numeric) |
| Job type | Employment type (categorical) (eg., Salaried, Self-employed, unemployed, business |
| Loan amount | Required loan amount (numeric) |
| Credit score | Credit Score (numeric) |

## 2. Data Preprocessing

Handle missing values
- fill with mean/median (numeric)
- fill with mode (categorical)

Remove duplicate records
- keep unique customer entries

Validate data types
- convert to numeric/categorical
- Correct inconsistent formats

Detect and treat outliers
- use IQR / z-score
- Cap or remove extreme values

## 3. Data Transformation Techniques

Encoding categorical variable (Job type)
- one hot encoding (creates binary columns)
- or label encoding (assigns numeric codes)

Normalization
- min-max scaling (0 to 1)

$$x' = (x - x_{min}) / (x_{max} - x_{min})$$

Standardization (feature scaling)
- converts to standard normal distribution $z = \frac{x - \mu}{\sigma}$

Feature Engineering (optional)
- create new features (e.g., income to loan)
- Transform skewed features (eg., log transf)

## 4. Transformed dataset

- All categorical data encoded.
- Numerical features normalized or standardized
- No missing or duplicate values
- consistent data format
- Ready for model training.

## 5. Data Ready for Analysis / Model Building

- cleaned and transformed customer data
- Numerical features scaled
- Categorical features encoded
- Sustainable for machine learning models (eg., Logistic Regression, DT, RF)
- used to Predict loan approval (Approved / Not Approved)

## 6. Applications / Outcomes

### Loan Approval Prediction
- Predict approved / rejected
- use classification models
- Improve decision accuracy

### Customer Risk Analysis
- Identify high-risk customers
- Analyze impact of income, Credit score, Job type
- Reduce default rates

### Business Insights
- understand customer Profiles
- feature importance analysis
- Support policy decisions

### Benefits
- faster and fairer loan approval
- Better risk management
- Improved customer satisfaction